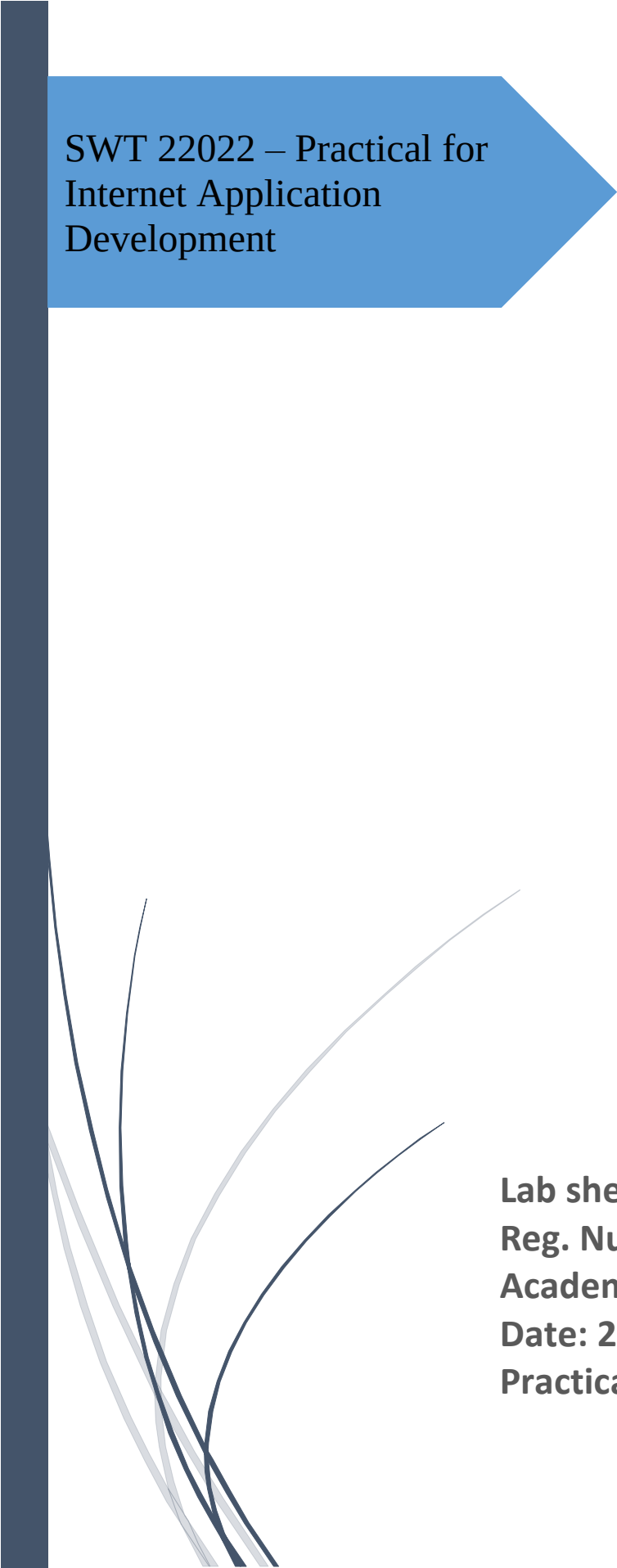




SWT 22022 – Practical for
Internet Application
Development

Department of Information
and Communication
Technology
Faculty of Technology



Lab sheet :14
Reg. Number: SEU/IS/20/ICT/084
Academic Year :2020/2021
Date: 2024.07.24
Practical No :15

Title:

- **HR Management System - 1**

Aims:

- **Create Laravel project**
- **Connect with database**
- **Implement APIs**

Tasks:

- **Create new project**
- **Start Laravel's local development server**
- **Create a database**
- **Add database to the project**
- **Run migrations**
- **Create a model**
- **Add variables to the model**
- **Update migrations**
- **Create a controller**
- **Implement CRUD operation**
- **Implement routes**
- **Test APIs**

Activities :

1. Create new project

composer create-project laravel/laravel hr-management-system

```
PS C:\xampp\htdocs\Lb14>
PS C:\xampp\htdocs\Lb14> composer create-project laravel/laravel hr-management-system
Creating a "laravel/laravel" project at "./hr-management-system"
Installing laravel/laravel (v11.3.3)
- Downloading laravel/laravel (v11.3.3)
- Installing laravel/laravel (v11.3.3): Extracting archive
Created project in C:\xampp\htdocs\Lb14\hr-management-system
> @php -r "file_exists('.env') || copy('.env.example', '.env');"
Loading composer repositories with package information
Updating dependencies
Lock file operations: 108 installs, 0 updates, 0 removals
- Locking brick/math (0.12.1)
- Locking carbonphp/carbon-doctrine-types (3.2.0)
- Locking dflydev/dot-access-data (v3.0.3)
- Locking doctrine/inflector (2.0.10)
- Locking doctrine/lexer (3.0.1)
- Locking dragonmantank/cron-expression (v3.4.0)
- Locking egulias/email-validator (4.0.2)
- Locking fakerphp/faker (v1.24.1)
- Locking filp/whoops (2.16.0)
- Locking fruitcake/php-cors (v1.3.0)
- Locking graham-campbell/result-type (v1.1.3)
- Locking guzzlehttp/guzzle (7.9.2)
- Locking guzzlehttp/promises (2.0.4)
- Locking guzzlehttp/psr7 (2.7.0)
- Locking guzzlehttp/uri-template (v1.0.3)
- Locking hamcrest/hamcrest-php (v2.0.1)
- Locking laravel/framework (v11.34.2)
- Locking laravel/pail (v1.2.1)
- Locking laravel/pint (v1.18.2)
```

2. Start Laravel's local development server

cd hr-management-system

php artisan serve

```
INFO Running migrations.

0001_01_01_000000_create_users_table ..... 203.38ms DONE
0001_01_01_000001_create_cache_table ..... 111.21ms DONE
0001_01_01_000002_create_jobs_table ..... 181.17ms DONE

PS C:\xampp\htdocs\Lb14> cd hr-management-system
PS C:\xampp\htdocs\Lb14\hr-management-system> php artisan serve
forking is not supported on this platform

INFO Server running on [http://127.0.0.1:8000].

Press Ctrl+C to stop the server
```

3. Create a database

Databases

Create database

hrsys utf8mb4_general_ci Create

4. Add database to the project

```
.env hr-management-system\.env
17 LOG_CHANNEL=stack
18 LOG_STACK=single
19 LOG_DEPRECATED_CHANNEL=null
20 LOG_LEVEL=debug
21
22 DB_CONNECTION=mysql
23 DB_HOST=127.0.0.1
24 DB_PORT=3306
25 DB_DATABASE=hrsys
26 DB_USERNAME=root
27 DB_PASSWORD=
28
29 SESSION_DRIVER=database
30 SESSION_LIFETIME=120
31 SESSION_ENCRYPT=false
32 SESSION_PATH=/
33 SESSION_DOMAIN=null
34
```

5. Run migrations

php artisan migrate

```
Creating migration table . 286.36ms DONE
INFO Running migrations.
0001_01_01_000000_create_users_table 3s DONE
0001_01_01_000001_create_cache_table 845.21ms DONE
0001_01_01_000002_create_jobs_table 1s DONE
PS C:\xampp\htdocs\Lb14\hr-management-system>
```

Table	Action	Rows	Type	Collation	Size	Overhead
cache	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 K1B	-
cache_locks	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 K1B	-
failed_jobs	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 K1B	-
jobs	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 K1B	-
job_batches	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 K1B	-
migrations	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_unicode_ci	16.0 K1B	-
password_reset_tokens	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 K1B	-
sessions	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 K1B	-
users	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 K1B	-
9 tables	Sum		3 InnoDB	utf8mb4_general_ci	224.0 K1B	0 B

6. Create a model

php artisan make:model Employee -m

```

PS C:\xampp\htdocs\Lb14\hr-management-system> php artisan make:model Employee -m

INFO: Preparing database.
Creating migration table . 286.36ms DONE

INFO: Running migrations.
0001_01_01_000000_create_users_table 3s DONE
0001_01_01_000001_create_cache_table 845.21ms DONE
0001_01_01_000002_create_jobs_table 1s DONE

PS C:\xampp\htdocs\Lb14\hr-management-system> php artisan make:model Employee -m

INFO: Model [C:\xampp\htdocs\Lb14\hr-management-system\app\Models\Employee-m.php] created successfully.

PS C:\xampp\htdocs\Lb14\hr-management-system> php artisan make:model Employee -m

INFO: Model [C:\xampp\htdocs\Lb14\hr-management-system\app\Models\Employee.php] created successfully.

INFO: Migration [C:\xampp\htdocs\Lb14\hr-management-system\database\migrations\2024_12_09_041519_create_employees_table.php] created successfully.

PS C:\xampp\htdocs\Lb14\hr-management-system>

```

7. Add variables to the Model

use Illuminate\Database\Eloquent\Model;

class Employee extends Model

{ use HasFactory;

protected \$fillable = [

'name',

'role',

'salary',

'age'

];

}

```

<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Employee extends Model
{
    use HasFactory;
    protected $fillable = [
        'name',
        'role',
        'salary',
        'age'
    ];
}

```

8. Update migrations

```

public function up(): void
{
    Schema::create('employees', function (Blueprint $table) {
        $table->id();
        $table->string('name')->nullable();
        $table->string('role')->nullable();
        $table->decimal('salary',10,2)->nullable();
        $table->integer('age')->nullable();
        $table->timestamps();
    });
}

```

php artisan migrate

```

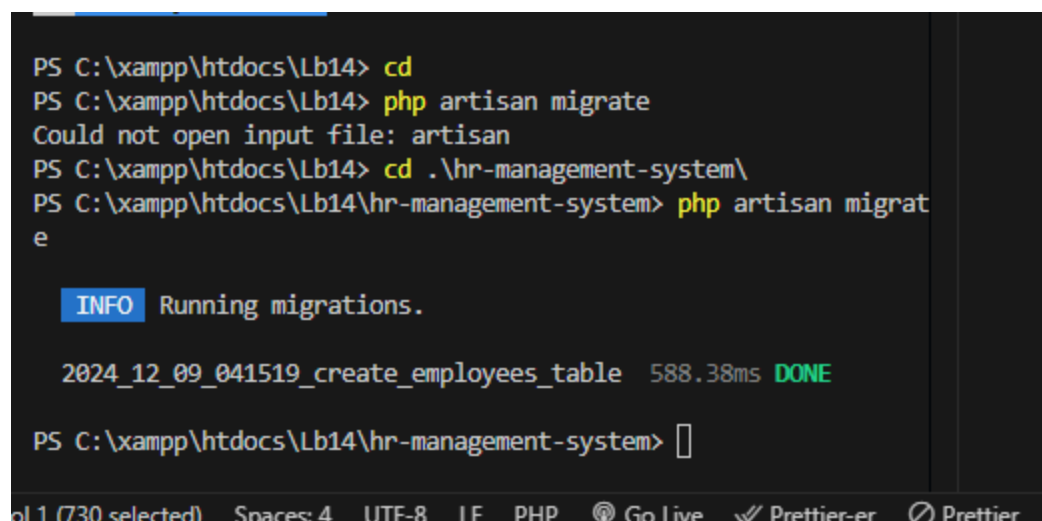
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void{
        Schema::create('employees', function (Blueprint $table) {
            $table->id();
            $table->string('name')->nullable();
            $table->string('role')->nullable();
            $table->decimal('salary',10,2)->nullable();
            $table->integer('age')->nullable();
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('employees');
    }
};

```



```

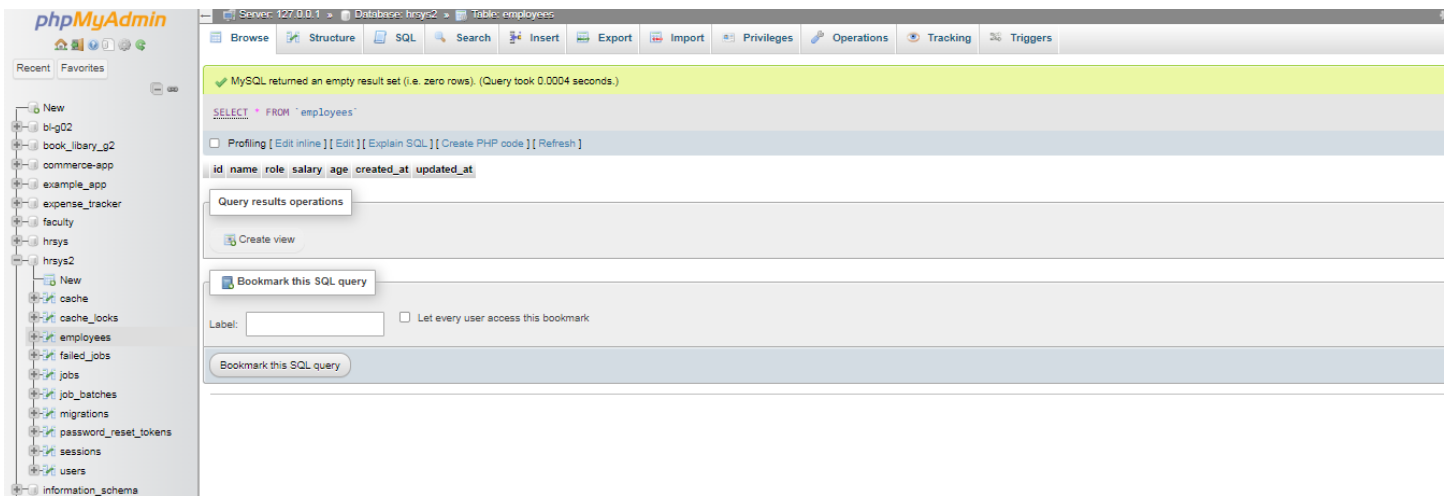
PS C:\xampp\htdocs\Lb14> cd
PS C:\xampp\htdocs\Lb14> php artisan migrate
Could not open input file: artisan
PS C:\xampp\htdocs\Lb14> cd .\hr-management-system\
PS C:\xampp\htdocs\Lb14\hr-management-system> php artisan migrate
e

INFO Running migrations.

2024_12_09_041519_create_employees_table 588.38ms DONE

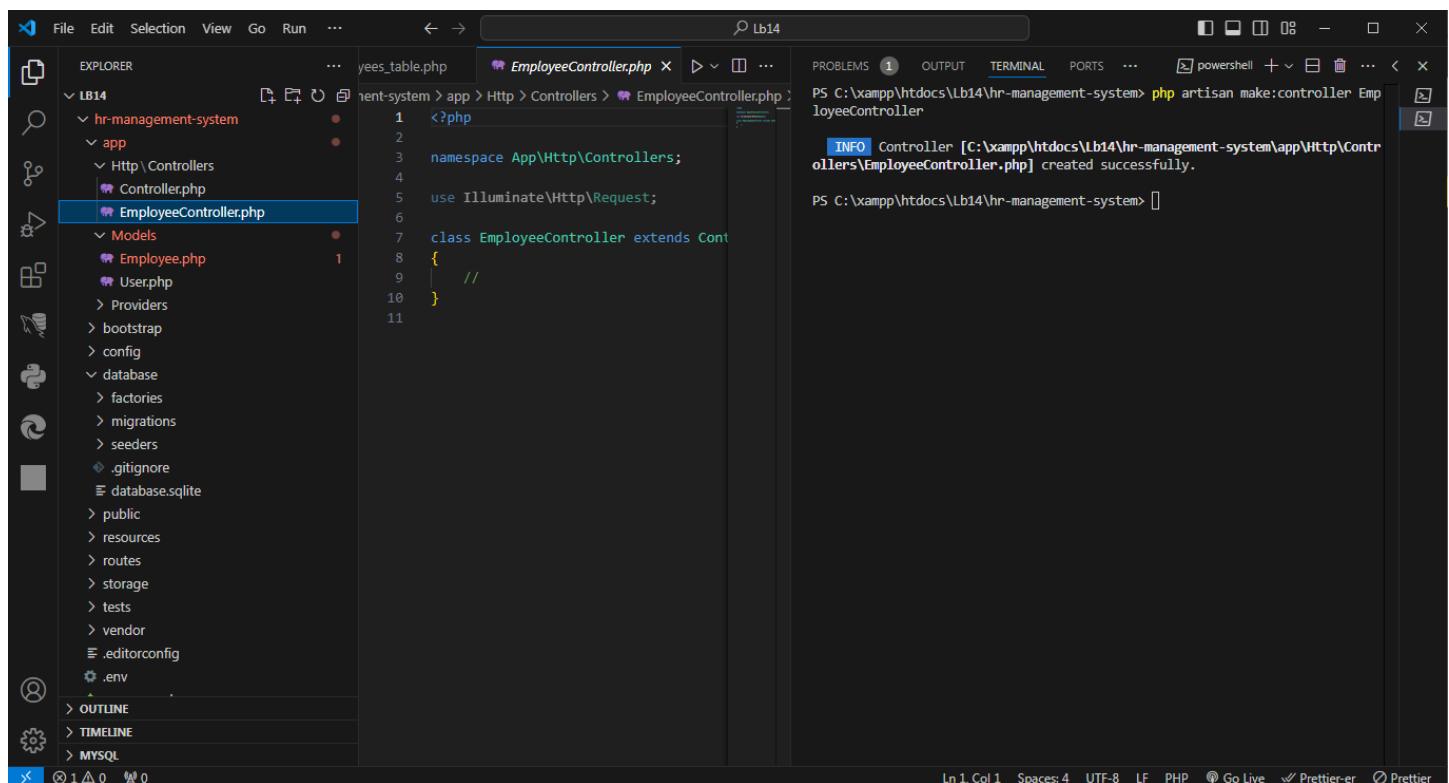
PS C:\xampp\htdocs\Lb14\hr-management-system> 

```



9. Create a controller

php artisan make:controller EmployeeController



10. Implement CRUD operation

```
public function store(StoreEmployeeRequest $request)
{
    Employee::create($request->validated());
}

public function all()
{
    return Employee::all();
}

public function get($id)
{
    return Employee::findOrFail($id);
}

public function edit($id)
{
    return Employee::findOrFail($id);
}

public function update(Request $request, $id)
{
    $employee = Employee::findOrFail($id);    return $employee->update($request->all());
}

public function delete($id)
{
    $employee = Employee::findOrFail($id);    return $employee->delete();
}
```

```
<?php

namespace App\Http\Controllers;

use App\Http\Requests\StoreEmployeeRequest;
use App\Models\Employee;
use Illuminate\Http\Request;

class EmployeeController extends Controller
{
    public function store(StoreEmployeeRequest $request){
        Employee::create($request->validated());
    }
    public function all(){
        return Employee::all();
    }
}
```

```

    public function get($id){
        return Employee::findOrFail($id);
    }
    public function edit($id){
        return Employee::findOrFail($id);
    }
    public function update(Request $request, $id)
    {
        $employee = Employee::findOrFail($id);
        return $employee->update($request->all());
    }
    public function delete($id){
        $employee = Employee::findOrFail($id);
        return $employee->delete();
    }
}

```

11. Implement routes

```

Route::get('/token', function () { return csrf_token(); });
Route::post('employee', [EmployeeController::class, 'store']);
Route::get('employee/all', [EmployeeController::class, 'all']);
Route::get('employee/{id}/edit', [EmployeeController::class, 'edit']);
Route::post('employee/{id}', [EmployeeController::class, 'update']);
Route::delete('employee/{id}', [EmployeeController::class,
'delete']);

```

```

<?php

use Illuminate\Support\Facades\Route;

Route::get('/', function () {
    return view('welcome');
});

Route::get('/token', function () {
    return csrf_token();
});

Route::post('employee', [EmployeeController::class, 'store']);
Route::get('employee/all', [EmployeeController::class, 'all']);
Route::get('employee/{id}/edit', [EmployeeController::class, 'edit']);

```

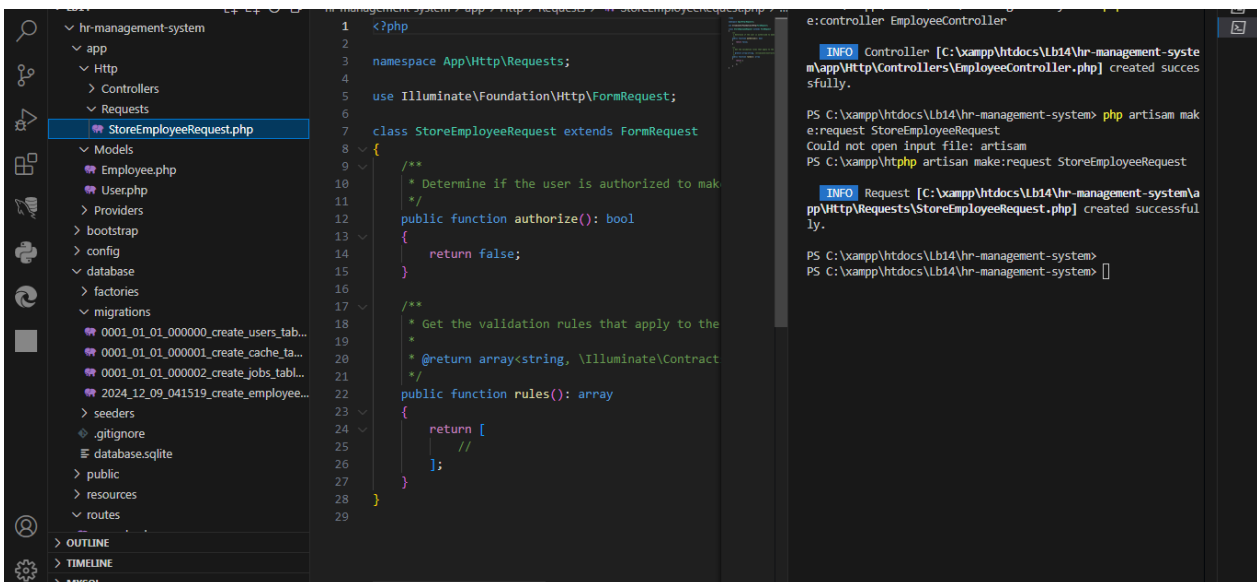
```
Route::post('employee/{id}', [EmployeeController::class, 'update']);
Route::delete('employee/{id}', [EmployeeController::class, 'delete']);
```

php artisan make:request StoreEmployeeRequest

```
Could not open input file: artisan
PS C:\xampp\htdocs\php artisan make:request StoreEmployeeRequest

INFO Request [C:\xampp\htdocs\Lb14\hr-management-system\app\Http\Requests\StoreEmployeeRequest.php] created successfully.

PS C:\xampp\htdocs\Lb14\hr-management-system>
PS C:\xampp\htdocs\Lb14\hr-management-system> |
```



12. Test APIs

Discussion:

- During this lab session, I gained hands-on experience in building a basic HR Management System using Laravel. The process began with creating a new Laravel project and starting the local development server to set up the working environment. I created and configured a database, connecting it to the Laravel project through the .env file, ensuring smooth communication between the application and the database. I then ran migrations to define the initial database schema and created a model to represent the employee data, adding the necessary attributes for the system. This was followed by updating migrations to include specific fields, reflecting real-world HR requirements like name, email, position, and salary. I created a controller to handle the business logic and implemented CRUD operations to manage employees effectively. Routes were defined to link API endpoints with controller actions, facilitating proper interaction between the client and server. Finally, I tested the APIs using tools like Postman, validating the functionality of each operation. This practical session helped me understand the end-to-end process of developing a Laravel-based application, reinforcing key concepts in database management, routing, and API implementation.